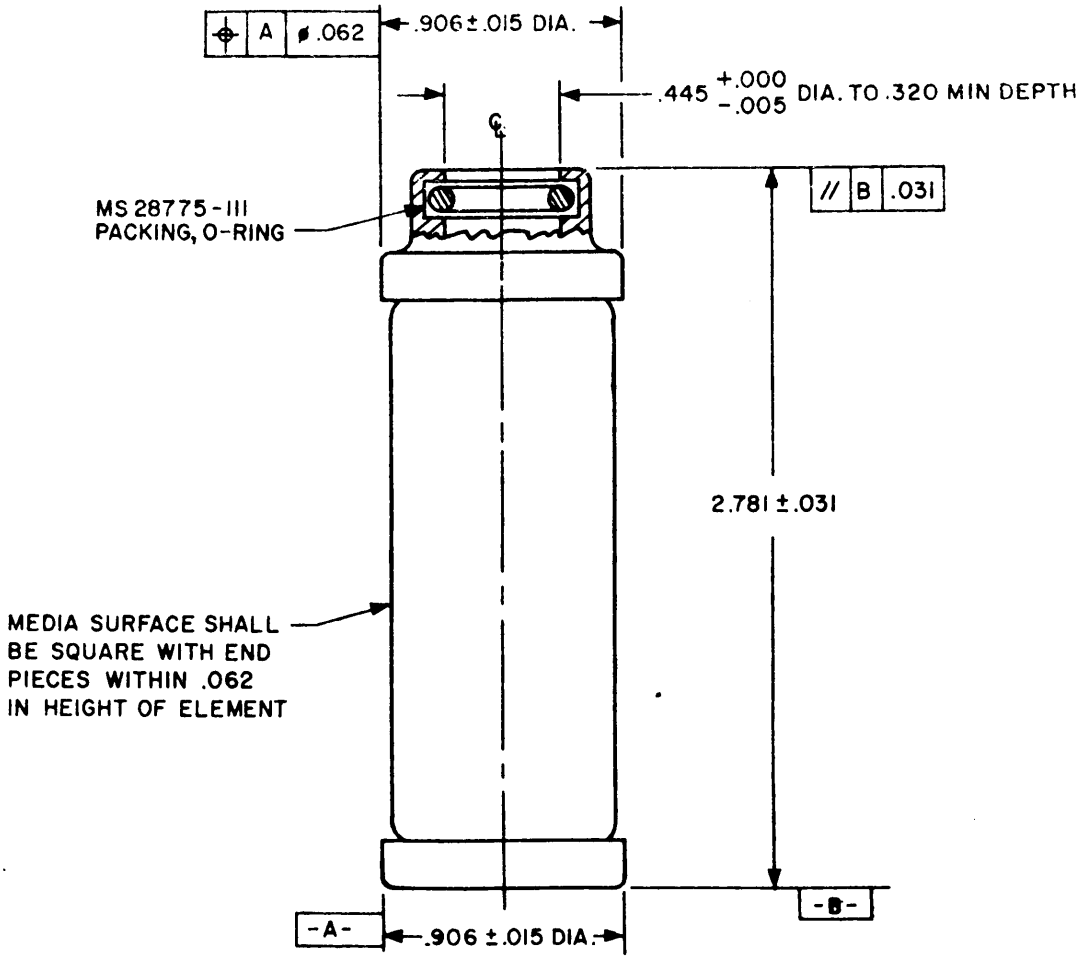
MIL-F-8815/20(AS)
14 October 1976

MILITARY SPECIFICATION SHEET

FILTER ELEMENTS, NONCLEANABLE , FLUID PRESSURE, HYDRAULIC LINE, 5 MICRON ABSOLUTE, A4 AIRCRAFT, 3.5 GPM

This specification is approved for use by the Naval Air Systems Command, Department of the Navy.

The complete requirements for procuring the filter elements described herein shall consist of this document and the latest issue in effect of Specification MIL-F-8815.



FSC 1650

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NOTES:

1. Dimensions in inches, unless otherwise specified. See above Figure.
2. Tolerances, decimals ±.010, angles ±l°.
3. Unless otherwise specified, surface roughness shall not exceed 125 microinches, per MIL-STD-10. Remove all burrs and break sharp edges .005 minimum.
4. Material shall be in accordance with the procurement specification.
5. Drawing symbols per ANSI 714.5 - 1973.

REQUIREMENTS:

Pressure Drop: Maximum pressure drop at rated flow across the filter element shall not exceed 10 psid.

Degree of Filtration: The filter element shall remove 100 percent of all contaminant larger than 5 microns and 95 percent of all test contaminant. The filter efficiency shall be a minimum of 95 percent.

Element Collapse: The filter element shall withstand a differential pressure of 2250 psid.

Cold Start: The element shall withstand 1500 psid differential pressure across the element for the flow cycles and fluid temperature as specified in the general specification. The element pressure drop shall not exceed 10 psid following the cold start test.

Maximum Particle Passed: The maximum particle passed shall be no greater than 5 microns,

Flow Fatigue: The filter element shall withstand the following flow fatigue pressure cycles at a cycling rate of 100 cpm maximum.

90,000 cycles at 20 psid pressure drop
8,000 cycles at 50 psid pressure drop
2,000 cycles at 100 psid pressure drop
100 cycles at 1500 psid pressure drop

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Part Number: The part number consists of the basic number of this specification sheet.

Example:

M8815/2O

TABLE I. CONTAMINANTS

Flow Rate (GPM)	Degree of Filtration Add Rate		Maximum Particle Add Rate		Pressure Buildup		Migration Analysis Value
	F-9 Beads (gms)	AC Dust (gms)	F-9 Beads (gms)	Carbonyl Iron E (gms)	Add Rate AC Dust (gms)	Minimum AC Dust to 90 psid (gms)	Maximum Allowable (gms)
3.5	0.070	0.070	0.020	0.010	0.2	1.0	0.002

Intended Use: The filter element covered by this specification sheet is intended for use in the model A4 aircraft operating with hydraulic fluid conforming to MIL-H-5606 at nominal operating pressures of 3,000 psi.

Preparing activity:
Navy - AS

(Project No. 1650-N052-12)

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-F-8815/20(AS) FILTER ELEMENTS, NONCLEANABLE, FLUID PRESSURE, HYDRAULIC LINE, 5 MICRON ABSOLUTE, A4 AIRCRAFT, 3.5 GPM

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

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